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OBA465 – Sports Analytics

Individual Assignment

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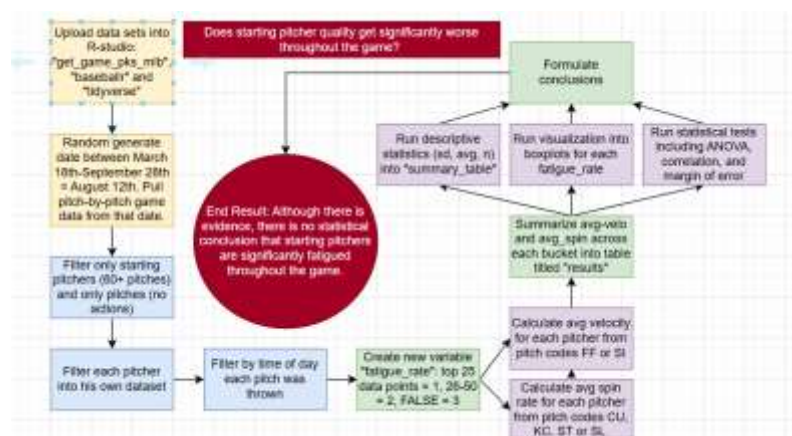
Starting Pitchers: Fatigue Throughout the Game

At what point does a starting pitcher become a liability rather than an asset? This report analyzes three pitch count thresholds (0-25, 26-50, 51+ pitch count) to determine whether there is a statistically significant relationship between pitch count and pitching effectiveness. This is broadened into determining if starting pitchers should be pulled from the game *sooner* for a better chance at winning the game through quality pitches.

To calculate pitcher effectiveness deep into ballgames, a date within the 2025 MLB season was randomly selected. August 12th, 2025, saw a sample of 30 starting pitchers take the bump, throwing from a range of 61 to 102 pitches. As mentioned prior, this report creates an independent variable calculating the pitch count bucket, denoted as the following: 1 = fresh, 2 = moderate, 3 = fatigued. Pitch count cutoffs were established from the sample, where the minimum observed pitch count was 61, providing sufficient observations in the fatigued category for pitch codes. Each pitcher threw either a four-seam fastball or a sinker, used to measure average velocity, along with a curveball, knuckle curve, slider, or sweeper, used to measure average spin rate.

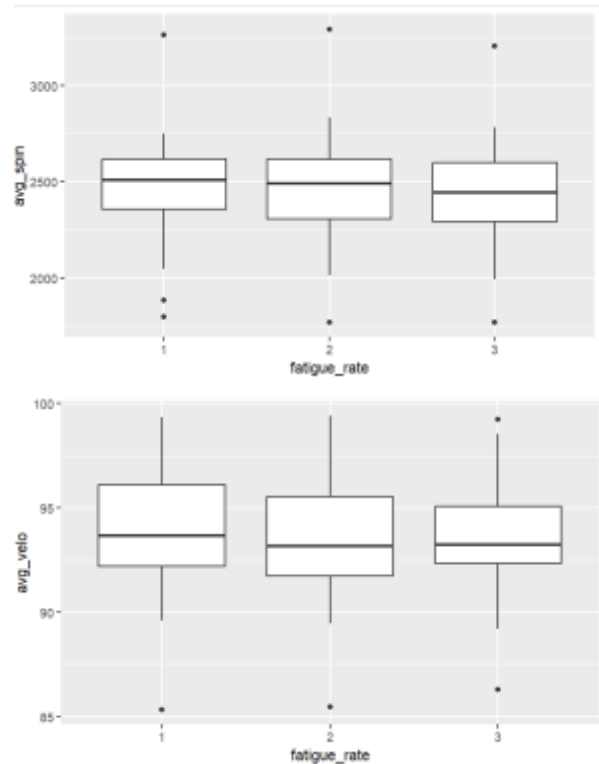
We are testing upon the basis that there is no change in pitcher performance as pitch count increases. Statistically significant observations for the alternative would suggest pitch quality declines enough to pull starting pitchers early in games to rely on the bullpen.

Statistical analyses in R-studio were conducted via the processes described in the flowchart (see right). Other tools used within R-studio include packages “dplyr” for calculations, “ggplot2” for visualization, and “baseballr” for velocity and spin rates.



The descriptive statistics and the boxplot provide a downward correlation between spin rates and velocity. In simple terms, there is evidence to suggest that our fatigue rates may impact on the pitcher throughout the game. However, there is no statistical basis surrounding this.

By running an ANOVA test, we are revealed to a large p-value of 0.8 for velocity, 0.931 for spin rates. These large p-values suggest there is little to no statistically significant evidence to support our fatigue thresholds truly impacting pitcher fatigue. In other words, although there are slight decreases in spin rates and velocity seen in the boxplots, pitchers aren't affected tremendously throughout the game based on the findings.



Although this project stays within its original parameters of questioning, it would be interesting to include another analytical variable to see hitter success on the pitchers throughout the game. A prime addition would be weighted on-base average (wOBA), batting average against the pitcher (BAA), or slugging percentage off the pitcher (SLG) each time through the order. This would add an extra element of “times through the lineup” where opposing batters will face the pitcher multiple times, possibly getting used to their pitching style. Instead of pitch count, we can mutate R-studio into finding each pitcher's success through the lineup once, twice, and three times seeing the batting order.